No wonder there is such a surge in interest in edge computing, and tech giants including Google, Microsoft, and Qualcomm have been upgrading their offerings to empower developers to scale and enable AI at the edge. Proper edge computing requires a seamless link between cloud, core, and edge. With thousands of hardware nodes to be managed, it is believed that DevOps will play an important role in this transition.

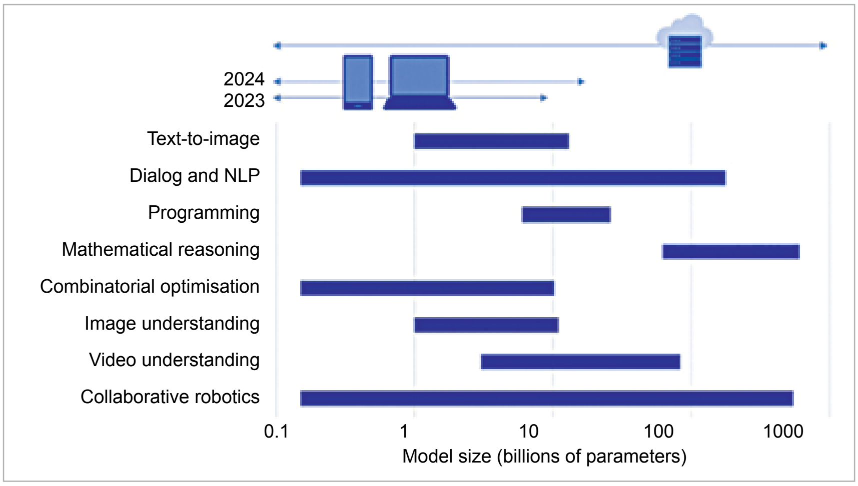
"There have been significant advances in edge computing. Many of the leading tech giants have brought in hardware architectural innovations, which provide increased computation, storage, and overall processing at the edge. Some of the hardware development boards are increasingly accessible from a cost perspective. Also, better connectivity solutions enabled by 5G bring faster communications between edge devices and enterprise systems," says Ajay Chaudhary, Associate Vice President - Engineering, GlobalLogic. "5G provides an average data transfer speed that is ten times faster than 4G, making it a key enabler of edge-based systems. This enhanced connectivity enables edge devices to communicate more quickly with enterprise systems, resulting in faster decision-making based on the data collected at the edge."

Many a change, here and there

Edge computing has already started bringing about changes—small and large—in fields ranging from shopping and ticketing to autonomous vehicles and Industry 4.0. "Amongst the emerging applications of edge computing, the most important ones include manufacturing, smart cities, healthcare, as well as autonomous vehicles and military operations," says Chaudhary.

Here is a small glimpse of what it can do for you...

- Irked by the time it takes to redirect you to the local store, when you visit a retail website? This can be avoided by using edge computing.



A significant number of generative AI models can run on the device, offloading the cloud (Source: Qualcomm Whitepaper "The Future of AI is Hybrid")

Location-aware edge computing devices will not only load the local site automatically but also show you relevant offers and customise the homepage with your preferences and shopping lists, without even a blink's delay!
- At large sporting events, the same ticket is sometimes scanned at multiple entrances, as it takes time for the scanned code to get updated in the cloud database. This scam can be avoided using scanners that communicate with each other at the edge. No network? No worries. Frauds will still be called out. Likewise, it can help reduce latency in almost every kind of security and surveillance solution.
- Data collection for marketing campaigns can eat into the bandwidth and cause delays, which are sometimes noticeable by the customer. With edge computing, marketing campaigns can be conducted smoothly without disturbing the customer experience.
- Edge computing can reduce latency and improve transaction times in the trading of non-fungible tokens (NFTs), otherwise considered a bandwidth-intensive operation.
- Sophisticated sensors can take corrective action on defects as soon as they are detected, without waiting for instructions from a centralised system, thereby improving quality control and saving time.
- Edge computing is an integral part of all Industry 4.0 efforts. "During the manufacturing process, certain autonomous processes require real-time decision-making to avoid any disruption in the production line. Edge devices are responsible for controlling these processes," explains Chaudhary. For example, an assembly line robot would place internal orders for a component, as soon as it realises that only a few are left in its stock pile.
- Medical devices can alert healthcare professionals in real time, based on built-in business rules, when a patient's condition requires immediate care. These would happen in true real-time fashion, without any dependency on the cloud platform. At times, this could be a life-saver, especially in remote areas where connectivity can be unreliable.
- Edge computing devices are the main components of military equipment, according to Chaudhary. "Edge devices have business rules for different battle conditions. These edge devices acquire data in real time and respond as per the battlefield